

Internship report: Modularity in Interactive Theorem Provers

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Interactive theorem provers (ITPs) do not easily allow users to adapt and extend existing codebases without either changing the original code or duplicating it. The pattern of extending definitions occurs particularly often in programming language theory, type theory, and program verification, leading to a lot of code duplication and high maintainance burden. Existing approaches to this problem either require encoding datatypes as complicated expressions, obfuscating the development, or rely on meta-programming facilities which were underdeveloped in most ITPs until recently.

We develop a principled way to produce modular code, leveraging the strengths of dependent types and modern meta-programming techniques.

***Note.** This research topic has been collaboratively developed by Arthur Adjedj and Yannick Forster. Arthur Adjedj was supervised by Yannick Forster and hosted in the Cambium team at the Centre Inria de Paris during a five-month internship.*

Introduction

AA: TODO some (most ?) of this should be put in related works rather than the intro. AA: Emphasis to be put on the fact formalisations can be extended after the fact, see STLC case-study. Interactive Theorem Provers (ITPs), also known as proof assistants, are tools which allow for the development and mechanical verification of formal proofs. ITPs can be used to provide a very high level of guarantees for software (in particular the complete absence of bugs), and several projects make use of proof assistants to verify real-world software, such as the CompCert C compiler (Leroy [2009]), the sel4 operating systems kernel (Lyons et al. [2025]), and AWS' authorization policy language, Cedar (AWS [2026]). They have also been used successfully to formalise landmark theorems in mathematics such as the Four Colour Theorem or the Feit Thompson theorem, and the use of proof assistants is on the rise in mathematics departments, with big mathematical developments like Lean's Mathlib library (The mathlib Community [2020]) getting more and more public attention.

Like other programming languages, ITPs fall victim to the the fact that reusing definitions can be non-trivial. This is generally referred to as the **expression problem** (Wadler [1998]):

“The goal is to define a datatype by cases, where one can add new cases to the datatype and new functions over the datatype, without recompiling existing code.”

In fact, reusing and adapting existing data structures and programs modularly is a challenge for which few solutions have been produced over the years in industrial programming languages (Oliveira and Cook [2012]). It is even worse for ITPs, where in addition to programs, one needs to adapt proofs and dependently typed programs as well. Most projects that try to extend existing formalisation resort to copying the original content and adapting it for its own purpose. Since proofs about programs are arguably even harder to maintain than programs, this copy-paste approach incurs a huge maintenance burden.

A central contribution on matters of modularity in regular programming languages, “Data Types à la Carte” (Swierstra [2008]), provides a simple and elegant solution to the expression problem based on a parametrical approach to modular syntax in Haskell. This solution, however, cannot be easily adapted to ITPs since ITPs needs to ensure consistency via their type system. Instead, existing approaches either rely on complex encodings of datatypes that leak to the user, or on meta-programming.

“Meta-Theory à la Carte” (Delaware et al. [2013]) and “Pyrosome” (Jamner et al. [2025]) base their modularity on internal encodings of types (namely, impredicative church-encodings in the former, and Generalized Algebraic Theories in the latter). In both cases, the constructions are inefficient, incapable of extending previously user-defined inductive types (*i.e.*, expecting users to rely on the aforementioned encodings from the ground up), and expose the underlying internals of the encodings to the user. Having to deal with encodings of types rather than types adds a heavy burden in particular for new users interested in program verification. The lesson to include inductive datatypes natively has been learned early by popular ITPs (namely Rocq and Lean), who at first had no primitive notions of inductive types in their systems and then resorted to add those. Nowadays, most systems usually possess a syntactic notion of (co-)inductive types, and justify the ability to define these via some classes of models, allowing users to work with the abstractions these types provide, rather than with their encoding in said models. The only popular system that still relies on encodings to define such types, and manages to hide their implementation details well, is Isabelle/HOL.

On the other hand, Rocq à la Carte (Forster and Stark [2020]) relies on the meta-programming capabilities offered by the MetaRocq Project (Sozeau et al. [2020]) to allow users to construct new inductive types and functions by merging other inductive types, and/or adding new constructors. New functions on a “merged” datatype can then be constructed by merging past functions. The metaprogram then uses the given piece of information to reconstruct a new inductive type, and new functions, based on the informations given by the user. While great for extending constructions “vertically” (*i.e.*, by adding constructors to a type), this system does not allow for horizontal extensions (*i.e.*, extending the type signature of inductive types and their constructors). Furthermore, this approach has been hindered in the past by the lack of good metaprogramming frameworks in ITPs.

None of these systems, independently of whether they use encodings or the meta-programming, handle all of the type-system of ITPs they are implemented for (*e.g.*, none handle co-inductive types), or allow users to extend past formalisations “after the fact”. Instead, formalisations have to be built from the ground up with the expectation that they will be extended with a specific framework in mind, making them much less useful for real-world uses.

Extensions

This section presents the technical details of the LeanALaCarte implementation. The main three contributions are 1. A system for partially mapping terms 2. a DSL command for defining an inductive type as an extension to another based on that produces the appropriate partial mappings between the two 3. a DSL for defining definitions/theorems as extensions of others, thus reusing the appropriate informations “after the fact”.

AA: TODO section intro AA: For each new thing presented, show the new syntax and how it can be used in practice. Build up an example over the sections using new tools progressively

Partial mappings

This section makes a partial attempt at justifying the implementation of partial mappings in our system, and why it can be trusted to make well-formed terms.

First, let’s define the syntax we’ll be using in this toy presentation. This is a basic dependently-typed system equipped with a predicative hierarchy of universes à la Russel, as well as constants and metavariables. For the sake of simplifying the presentation, we will assume constants cannot be unfolded (*i.e.* we will only care about β/η -reduction, not δ -reduction). This, in particular, does not give good justification as to why recursors of inductive types may be extended safely.

Terms : $t, f, A, B ::= x$	(variable)
d	(constant)
m	(metavariable)
$f t$	(application)
$\lambda x : A. t$	(abstraction)
$(x : A) \rightarrow B$	(Π -type)
$\mathcal{U}_i$	(universe)

Figure 1: Syntax of our type theory

The usual typing judgements one would expect apply. These judgement need to carry not only the usual variable context Γ , but also a global constants context Σ to handle the type of constants, similarly to Sozeau et al. [2025], as well as a metavariable context Θ that holds, for each metavariable, both the context and the type of said metavariable, similarly to Kovács [2020]. The idea is that constants may be mapped to some term containing “holes”, i.e metavariables, allowing us to make the partial maps.

We define a judgement $\Sigma \mid \Phi \mid \Theta \mid \Gamma \Rightarrow \Delta \vdash t \Rightarrow t' : A \Rightarrow A'$ encompassing the behaviour of the partial mapping in practice. This judgement carries the same contexts found in the aforementioned typing judgements, as well as a mapping context Φ , which maps constants to the bundling of a metavariable context and a term that may contain the metavariables present in the context it's bundled with. The judgement states tracks both the base variable context and a variable context for the partially mapped term, as well as the type of both the original term and the partially mapped one. The partial mapping acts structurally over the usual constructors of our type-theory, and relies on the mapping context to translate constants into their partial mapping. Furthermore, when partially mapping a constant, the metavariable context it carries is weakened/lifted, such that every metavariable lives in some telescope over translated variable context.

$$\boxed{\Sigma \mid \Phi \mid \Theta \mid \Gamma \Rightarrow \Delta \vdash t \Rightarrow t' : A \Rightarrow A'}$$

(In global context Σ , mapping context Φ , metavariable context Θ , term t of type A in context Γ maps to term t' of type A' in context Δ)

$$\begin{array}{c}
\frac{}{\Sigma \mid \Phi \mid \Theta \mid \Gamma \Rightarrow \Delta \vdash \mathcal{U}_i \Rightarrow \mathcal{U}_i : \mathcal{U}_{i+1} \Rightarrow \mathcal{U}_{i+1}} \quad \frac{(x : A) \in \Gamma \quad (x : A') \in \Delta}{\Sigma \mid \Phi \mid \Theta \mid \Gamma \Rightarrow \Delta \vdash x \Rightarrow x : A \Rightarrow A'} \\
\\
\frac{((d : A) \mapsto (\Theta, t : A')) \in \Phi}{\Sigma \mid \Phi \mid \uparrow_{\Delta} \Theta \mid \Gamma \Rightarrow \Delta \vdash d \Rightarrow t : A \Rightarrow A'} \\
\\
\frac{\Sigma \mid \Phi \mid \Theta_1 \mid \Gamma \Rightarrow \Delta \vdash f \Rightarrow f' : ((x : A) \rightarrow B) \Rightarrow ((x : A') \rightarrow B') \quad \Sigma \mid \Phi \mid \Theta_2 \mid \Gamma \Rightarrow \Delta \vdash t \Rightarrow t' : A \Rightarrow A'}{\Sigma \mid \Phi \mid \Theta_1 \cup \Theta_2 \mid \Gamma \Rightarrow \Delta \vdash f t \Rightarrow f' t' : B[x := t] \Rightarrow B'[x := t']} \text{MV}(\Theta_1) \cap \text{MV}(\Theta_2) = \emptyset \\
\\
\frac{\Sigma \mid \Phi \mid \Theta \mid (\Gamma, x : A) \Rightarrow (\Delta, x : A') \vdash t \Rightarrow t' : B \Rightarrow B'}{\Sigma \mid \Phi \mid \Theta \mid \Gamma \Rightarrow \Delta \vdash (\lambda x : A. t) \Rightarrow (\lambda x : A'. t') : ((x : A) \rightarrow B) \Rightarrow ((x : A') \rightarrow B')} \\
\\
\frac{\Sigma \mid \Phi \mid \Theta \mid (\Gamma, x : A) \Rightarrow (\Delta, x : A') \vdash t \Rightarrow t' : B \Rightarrow B'}{\Sigma \mid \Phi \mid \Theta \mid \Gamma \Rightarrow \Delta \vdash ((x : A) \rightarrow B) \Rightarrow ((x : A') \rightarrow B') : \mathcal{U}_i \Rightarrow \mathcal{U}_i}
\end{array}$$

Figure 2: Judgement rules for mappings

The data contained in this judgement is enough to ensure any partially mapped term to be well-typed. Note that this theorem critically relies on the fact that this system only uses

Theorem 1 If for all $((d : A) \mapsto (\Theta, t : A')) \in \Phi$, we have (1) $\Sigma \mid \varepsilon \mid \varepsilon \vdash d : A$, (2) $\Sigma \mid \Theta \mid \varepsilon \vdash t : A'$ and (3) $\Sigma \mid \Phi \mid \Theta \mid \varepsilon \Rightarrow \varepsilon \vdash A \Rightarrow A' : \mathcal{U}_i \Rightarrow \mathcal{U}_i$ for some i , then for all Γ, t, A s.t $\Sigma \mid \Gamma \vdash t : A$, there exists Δ, t', A' s.t $\Sigma \mid \Phi \mid \Theta \mid \Delta \vdash t' : A'$ and $\Sigma \mid \Phi \mid \Theta \mid \Gamma \Rightarrow \Delta \vdash t \Rightarrow t' : A \Rightarrow A'$

Proof: by induction on the typing judgement $\Sigma \mid \Gamma \vdash t : A$. AA: TODO: actually prove this 😊

Inductive extensions

With this framework constructed, we may now use it to construct practical partial mappings, by producing useful mapping contexts. Since most constants in proof assistants are either inductive constructions (i.e an inductive type, a constructor or a recursor), or a declaration (i.e a definition or theorem), we first focus on inductive extensions, and then look at how they can be used to construct useful declaration extensions.

First, let us look at the shape of an inductive type. Inductive types are shaped as follows:

inductive Ind $[(p_j : P_j)]_j : [(i_k : I_k)]_k \rightarrow \mathcal{U}_n$ **where**

| **ctor**₁ : $[(a_{1,l} : A_{1,l})]_l \rightarrow I [p_j]_j [d_{1,k}]_k$

...

| **ctor**_n : $[(a_{n,l} : A_{n,l})]_l \rightarrow I [p_j]_j [d_{n,k}]_k$

An inductive (family of) type(s) is composed of a number of different components: A list of parameters which constitute a context, a telescope of indices over that context of parameters, and a list of constructor. Constructor contain a telescope of fields living over the parameter context, and instantiate the indices of the inductive type they inhabit in a context containing those fields. Basic examples of inductive types include the natural numbers, list, and vectors (i.e list indexed by their lengths):

```
inductive Nat : where   inductive List (A : )
| zero : Nat
| succ : Nat -> nat
```

AA: Mention all the various auxiliary declarations that Lean uses internally and that need to be mapped appropriately

Definition extensions

Extending matchers

AA: Explain that the internal implementation of matchers does not reflect the kind of extension you would want to see in practice, mention auto-generalizations, and how they are actually handled in practice.

Making extensions modular

AA: TODO section intro

Inductive modularity

Definition modularity

AA: Empty for now, no work has been done here, write down your ideas

Case study: extending STLC

Related works

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